

River Samet

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EDUCATION

University of California, Los Angeles (UCLA)

Graduated March 2026

Master of Science — Chemistry, computational emphasis | 3.89 GPA

University of California, Santa Barbara (UCSB)

Graduated June 2023

Bachelor of Science — Chemistry | 3.96 GPA

DATA SCIENCE/ANALYTICS AND RESEARCH EXPERIENCE

Graduate Student Researcher

October 2024 – Present

Houk and Gutierrez Groups | University of California, Los Angeles

- Build 20+ Python and Bash pipelines to automate large-scale computational workflows, improving efficiency and reproducibility
- Apply machine learning, statistical modeling, and quantum mechanical calculations to analyze and predict 10+ different complex organic, organometallic, and biological reaction mechanisms
- Process and analyze molecular simulation datasets containing thousands of entries to extract quantitative insights into chemical and biological reaction dynamics
- Present technical insights weekly to interdisciplinary teams, translating complex analyses into clear conclusions
- Communicate complex technical concepts to large undergraduate audiences across five different chemistry courses while employed as a teaching assistant

Computational Chemistry Researcher

June 2023 – September 2024

Aue Group | University of California, Santa Barbara

- Developed workflows using Linux command-line tools to analyze multi-outcome reaction systems (5+ pathways), requiring complex data interpretation and classification
- Managed high-throughput execution and analysis of thousands of reaction simulation jobs using automated Bash scripting
- Identified five different novel organic reaction pathways through data-driven analysis and literature-informed modeling

Organic Chemistry Research Intern

October 2022 – January 2023

Zhang Group | University of California, Santa Barbara

- Analyzed experimental data (yield, purity, structure) to evaluate organic synthetic process efficiency and scalability
- Optimized reaction conditions using data-driven decision-making across cost, energy, safety, and performance metrics

Laboratory Research Intern and Technician

May 2022 – October 2022

Reaction35, LLC | Goleta, CA

- Analyzed salt solubility and ion chromatography data in Microsoft Excel to identify trends and evaluate experimental outcomes
- Presented results in weekly meetings to technical and executive stakeholders to support data-informed process development

Chemical Engineering Research Intern

October 2021 – June 2022

McFarland Group | University of California, Santa Barbara

- Developed analytical methods using Microsoft Excel and Mathematica to extract quantitative kinetic insights from complex datasets with 10,000+ entries related to molten metal-catalyzed methane pyrolysis
- Presented technical results in weekly group meetings to interdisciplinary faculty, staff, and students

SKILLS

Programming & Relational Databases: Python (pandas, NumPy), SQL (PostgreSQL, SQLAlchemy), Bash, Excel | **Machine Learning:** scikit-learn, TensorFlow, Keras, transfer learning, data augmentation, permutation feature importance | **LLMs & GenAI:** RAG, fine-tuning (LoRA, distillation), Hugging Face Transformers, sentence-transformers, vector search (sqlite-vec), agentic workflows | **Data Visualization:** Tableau, Matplotlib, seaborn | **Cloud & HPC:** AWS (S3, SageMaker), Linux, HPC clusters | **Statistical Analysis:** hypothesis testing (A/B testing, t-tests), regression | **Computational Chemistry:** Amber, Gaussian, ORCA, pytraj, MDTraj, RDKit, PyMOL, Chimera, GaussView

SELECTED DATA SCIENCE PROJECTS

QTRAP (Quantum mechanics/molecular mechanics Trajectory Running, Analysis, and Preprocessing)

Houk Group | University of California, Los Angeles

- Developed the first-ever publicly available end-to-end quasiclassical QM/MM dynamics workflow (Python and Bash source code)
- Built permission-scoped tools and documentation that let an AI agent run the full workflow on a remote HPC cluster with minimal input
- Reduced hands-on time required to preprocess, run, and analyze simulations from weeks to just hours

AI Procedural Assistant: RAG With LLM Fine-Tuning

Volunteer Project for WALL-OP WATER | Garden Grove, CA

- Built an offline-capable RAG assistant that answers employee questions from company SOPs and related in-domain material within a secure local environment, reducing time supervisors spend on procedural questions
- Retrieval: sqlite-vec vector store with multi-qa-MiniLM-L6-cos-v1 embeddings (Recall@6 = 0.90); generation: Qwen2.5-1.5B fine-tuned by knowledge distillation with LoRA (Hugging Face PEFT/TRL) on AWS SageMaker with data staged in S3
- Best fine-tuned model reached over 80% accuracy on a 55-question held-out SOP test set with minimal hallucination and over-refusal, and 90% accuracy on out-of-SOP in-domain questions (vs. 45% for the base model)

CNN Image Classification

Independent Project

- Compared custom CNNs (with/without data augmentation) and MobileNetV2 transfer learning on 6,164 traffic sign images (58 classes)
- Fine-tuned MobileNetV2 reached ~98% test accuracy vs. 96% for the best custom CNN; packaged the final model as a command-line tool that reports predictions and confidence for user-supplied images